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HACKATHON IDEAS GUIDE

VIBE CODING BUILDERS HACKATHON

10 battle-tested project ideas powered by the last30days research skill. From signal scanning to micro-SaaS, build what people actually want.

• 10 IDEAS • 16+ DATA SOURCES • 30-DAY LAUNCH

30D

POWERED BY LAST30DAYS-SKILL V3.3.2 •
GITHUB.COM/ROMAN-RYZENADVANCED/LAST30DAYS-SKILL

01 // WHY THIS GUIDE

Build What People Actually Want

Stop guessing. Use real engagement signals from 16+ platforms to validate your hackathon project before you write a single line of code.

16+

DATA SOURCES

1,012

TESTS PASSING

30d

LAUNCH WINDOW

What is the last30days-skill?

A two-in-one AI skill that researches what people actually say about any topic across Reddit, X/Twitter, YouTube, TikTok, Instagram, Hacker News, Polymarket, GitHub, Bluesky, TruthSocial, Threads, Pinterest, Digg, Perplexity, Xiaohongshu, and the web. It scores results by real engagement (upvotes, likes, views, real money odds) and synthesizes everything into one grounded brief. The business launch system then finds validated opportunities from those signals and generates everything you need to launch in 30 days or less.

ENGINE

Multi-source parallel search

SCORING

Engagement-weighted signals

OUTPUT

Ranked evidence clusters

LAUNCH

30-day business timeline

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02 // PROJECT IDEAS

10 Hackathon-Worthy Builds

Each idea leverages the last30days-skill's research engine to find real demand before you build.

IDEA 01

BEGINNER

Trend-to-App Builder

Use the signal scanner to find trending pain points across Reddit, X, and TikTok, then auto-generate a vibe-coded micro-app that solves the problem. The research engine surfaces what people are actively complaining about; your app addresses it. Think "lazy dev meets real-time demand."

last30days

Next.js

z.ai GLM

Vercel

IDEA 02

INTERMEDIATE

Competitor Intelligence Dashboard

Build a real-time A/B/C comparison tool using the comparison mode. Feed it "Tool A vs Tool B vs Tool C" and get head-to-head tables with engagement scores from every platform. Add a watchlist that monitors for changes and delivers daily Slack digests when sentiment shifts. Perfect for indie hackers tracking their niche.

comparison-mode

watchlist

React

Slack API

IDEA 03

BEGINNER

Developer Pulse Newsletter

Auto-generate a weekly "Dev Pulse" digest by scanning Hacker News, GitHub trending repos, and Reddit r/programming with the briefing engine. Surface the top 5 stories, 3 emerging tools, and 1 controversial take each week. Use HTML brief output for beautiful email-ready formatting. Zero manual curation.

briefing.py

watchlist.py

HTML briefs

Mailchimp

IDEA 04

ADVANCED

Niche Discovery Bot for Indie Hackers

Build a conversational bot that guides indie hackers through the full signal-scan-to-business workflow. The bot uses the interactive gate pattern: it presents scored opportunities, waits for the user to

02 // PROJECT IDEAS (CONTINUED)

IDEA 05

INTERMEDIATE

Micro-SaaS Launch Pad

Combine the signal scanner with the landing page factory and ad launch playbook into a one-click SaaS launcher. Pick a niche from the scored list, and the system generates a landing page, calculates unit economics, creates an ad plan, and builds your 30-day timeline automatically. From idea to live in 24 hours.

signal-scanner

landing-page-factory

ad-playbook

timeline-system

IDEA 06

BEGINNER

Social Signal Validator

Validate any hackathon idea in 60 seconds. Paste your concept, and the tool searches Reddit, X, and HN for evidence that people want it. Returns a "Signal Score" (0-10) with the actual quotes and engagement metrics that prove demand exists or doesn't. Kill bad ideas fast before you waste your hackathon weekend.

quick-scan

engagement-scoring

Streamlit

Python

IDEA 07

ADVANCED

Pivot Predictor

Turn the failure/pivot protocol into an interactive diagnostic tool. Connect it to your live product metrics (ad CTR, conversion rate, bounce rate) and it auto-detects which of the 6 failure modes you are heading toward, then prescribes exact pivot actions with timelines. "If CTR less than 1% after 7 days, switch to community-led growth."

failure-pivot

signal-scanner

Analytics API

React

IDEA 08

INTERMEDIATE

Provider Marketplace Finder

Use data reconnaissance + provider sourcing to build a hyperlocal marketplace finder. Enter your city and service type (cleaning, tutoring, repairs) and the tool builds a real market map with contactable providers, pricing intelligence, and quality signals. Think "Airbnb for gig workers" powered by real Reddit and Facebook data.

data-recon

provider-sourcing

local-channel-mapper

Next.js

02 // PROJECT IDEAS (CONTINUED)

IDEA 09

ADVANCED

AI Research Agent for Hackathons

Build an autonomous research agent that uses agent mode to investigate hackathon topics end-to-end. Give it a theme like "fintech for Gen Z" and it resolves key entities, searches all 16+ sources in parallel, clusters cross-source stories, ranks by engagement, and produces a shareable HTML brief with citations. Teammates get the brief in Slack before the hackathon starts.

agent-mode

entity-resolution

cluster-merge

HTML briefs

IDEA 10

INTERMEDIATE

Vibe Coder Revenue Calculator

A beautiful web tool that wraps break_even.py and niche_validator.py into a visual calculator. Enter your niche, location, and budget. Get break-even volume, margin analysis, runway projection, and a 12-criteria viability score. Pair it with real-time signal data from the research engine so the numbers are backed by actual market signals, not guesses.

break_even.py

niche_validator.py

signal_score.py

Next.js

03 // DATA SOURCES

16+ Platforms at Your Fingertips

SOURCE	SIGNAL TYPE	AUTH
Reddit	Top comments + upvotes	Free
X / Twitter	Threads, reactions, hot takes	Cookies/API
YouTube	Transcripts + top comments	yt-dlp
TikTok	Viral engagement signals	ScrapeCreators
Hacker News	Developer consensus	Free
Polymarket	Real-money odds	Free
GitHub	Stars, PRs, release velocity	Free
Instagram	Reels + transcripts	ScrapeCreators

04 // GETTING STARTED

Use the 30-Day Skill on chat.z.ai

Get unlimited hackathon ideas by chatting with the Z.ai agent using the last30days skill from GitHub.

1 Install the Skill

Open your terminal and run the skill installer for your agent host:

```
# Install for any agent host (Claude Code, Codex, Cursor, etc.)
npx skills add roman-ryzenadvanced/last30days-skill -g

# Or install standalone
git clone https://github.com/roman-ryzenadvanced/last30days-skill.git
```

2 Use on chat.z.ai

Once installed, open chat.z.ai and invoke the skill with natural language prompts. The agent will research across 16+ platforms and return scored results.

```
# Research any topic
/last30days vibe coding tools

# Compare tools head-to-head
/last30days Cursor vs Windsurf vs Bolt

# Auto-discover competitors
/last30days "AI code editors" --competitors

# Get business ideas from signals
/last30days "micro-SaaS for developers"

# Quick scan for opportunities
/last30days "indie hacker pain points" --emit=compact
```

3 Explore Prompt Variants

The skill ships with 5 specialized prompt modes for different use cases:

QUICK SCAN

Top 5 opportunities in 5 min

DEEP DIVE

Single-niche full analysis

PIVOT

Diagnose and fix failing ideas

SCALE

Growth path for working ideas

04b // AGENT MODE

Use Agent Mode on chat.z.ai

No local install needed. Just tell the Z.ai agent to use the skill — it runs autonomously.

The "Please use this skill" Pattern

The easiest way to use the last30days skill is through **chat.z.ai**. No terminal, no install, no configuration. Just paste this pattern:

```
/* The magic pattern — just paste this into chat.z.ai */  
Please use this skill https://github.com/roman-ryzenadvanced/last30days-skill to find  
hackathon ideas about AI developer tools
```

```
/* The agent does everything automatically */  
// 1. Loads the skill from the GitHub URL  
// 2. Runs Agent mode autonomously  
// 3. Returns: Key Findings + What I learned + Stats
```

1 Copy the Pattern

Replace the topic after "to" with whatever you want to research.

```
/* Find hackathon ideas */  
Please use this skill https://github.com/roman-ryzenadvanced/last30days-skill to find  
hackathon ideas about vibe coding tools  
  
/* Compare tools */  
Please use this skill https://github.com/roman-ryzenadvanced/last30days-skill to compare  
Cursor vs Windsurf vs Bolt for hackathon builders  
  
/* Validate demand */  
Please use this skill https://github.com/roman-ryzenadvanced/last30days-skill to validate if  
there is real demand for an AI code review tool with fewer false positives  
  
/* Research pain points */  
Please use this skill https://github.com/roman-ryzenadvanced/last30days-skill to research  
what indie hackers are struggling with in 2026 and find hackathon-worthy problems
```

2 How It Works

When you paste the pattern, the Z.ai agent automatically loads the skill and runs Agent mode. Here's the pipeline:

LOAD

DECOMPOSE

Skill loaded from GitHub URL

Question split into sub-queries

SEARCH

16+ sources scanned in parallel

SCORE

Results ranked by engagement

MERGE

Cross-source clusters combined

REPORT

Key Findings + synthesis + stats

Pro Tip: Be Specific

The more specific your prompt, the better the research. Instead of "find ideas about AI," say "find hackathon ideas about AI tools that solve developer pain points with code review." **Specificity = better entity resolution = higher signal.**

05 // THE BUILD WORKFLOW

From Signal to Shipping

Follow this proven 5-step workflow to go from research to a live product in 30 days.

Scan Signals

→

Choose Niche

→

Data Recon

→

Find Providers

→

Launch & Test

Hackathon Weekend Cheat Sheet

Day 1 (Friday Night): Run `/last30days "your hackathon theme" --emit=compact` to find the highest-signal idea. Use `--competitors` to see what already exists. Pick your niche at the interactive gate.

Day 2 (Saturday): Build the core feature. Use the landing page factory for your demo page. Run the unit economics calculator to prove viability. Generate your 30-day timeline.

Day 3 (Sunday): Polish the demo, create a shareable HTML brief of your research findings, and present with real engagement data backing every design decision.

06 // PRO TIPS

Winning Strategies

Show Your Research

Judges love data-backed decisions. Use `--emit=html` to generate a shareable HTML brief showing exactly what Reddit, X, and HN users are saying about the problem you solve. Real quotes with engagement metrics beat slides every time.

Use Comparison Mode

Run `/last30days "Your App vs Competitor A vs Competitor B"` to produce a head-to-head comparison table. This shows judges you understand the landscape and have a genuine gap to exploit.

Watchlist for Live Demos

Set up `watchlist.py` before the hackathon so you can show live trend monitoring during your demo.
"This alert came in 2 hours ago: sentiment on X shifted 15% positive about AI code editors." Real-time data in a demo is unforgettable.

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07 // QUICK REFERENCE

Skill Commands Reference

```
# Basic research
/last30days "topic here"

# Compact output (fast)
/last30days "topic" --emit=compact

# Full HTML brief
/last30days "topic" --emit=html

# Compare 3+ entities
/last30days "A vs B vs C"

# Auto-discover competitors
/last30days "topic" --competitors

# Agent mode (autonomous)
/last30days "topic" --agent

# Agent mode on chat.z.ai (no install needed!)
Please use this skill https://github.com/roman-ryzenadvanced/last30days-skill to research your topic

# ELI5 mode
/last30days "topic" --eli5
```

08 // KEY LINKS

Essential Resources

RESOURCE	URL
GitHub Repo	github.com/roman-ryzenadvanced/last30days-skill
Z.ai Chat	chat.z.ai
Z.ai Coding Plans	z.ai/subscribe?ic=ROK78RJKNW
rommark.dev	rommark.dev
Skill Install	<code>npx skills add roman-ryzenadvanced/last30days-skill -g</code>
Config Guide	CONFIGURATION.md (in repo)

Ready to Build?

Install the `last30days` skill, fire up `chat.z.ai`, and let real engagement data guide your next hackathon project.

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The Philosophy

Engagement score > search ranking. Build where the conversation is happening.

Social relevancy, not SEO relevancy. Upvotes and likes, not keyword matches.

Start messy, iterate fast. \$50 is enough to start.

Collaboration over generation. You decide, the AI executes.

The interactive gate is sacred. Never skip it, never auto-pick.